

CSE 331: Computer Security Fundamentals

Spring 2026

Web Security:
Tracking & Privacy

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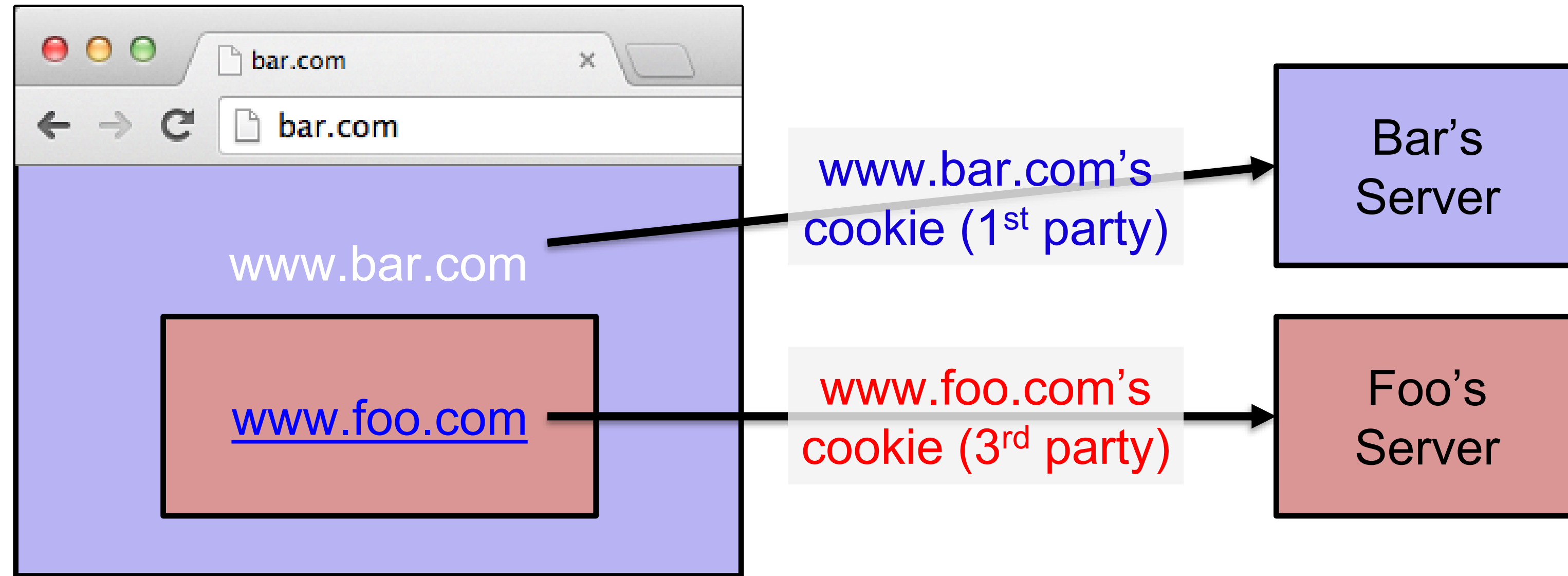


Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

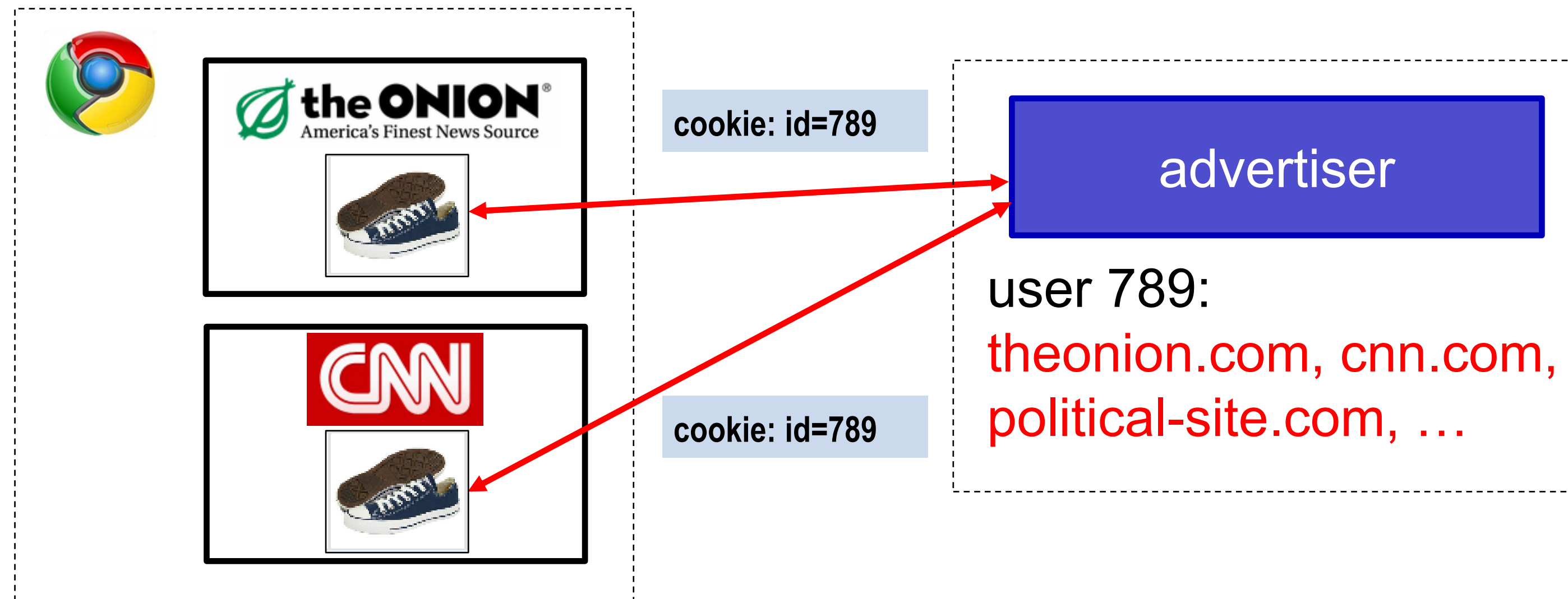
First and Third-Party Cookies

- **First-party cookie:** belongs to top-level domain.
- **Third-party cookie:** belongs to domain of embedded content (such as image, iframe).



Anonymous Tracking

Trackers included in other sites use **third-party cookies** containing unique identifiers to create browsing profiles.



Anonymous Tracking Example (1)

- **Action:** You visit a first-party site (e.g., CNN.com).
- **Mechanism:** Inside CNN's html code, there is an invisible <iframe> or a tiny 1x1 pixel image pointing to rubiconproject.com.
They are called **web beacons** or **tracking pixels**.
- **Result:** Because your browser is sending a request to Rubicon, it can send a Set-Cookie header. This cookie contains a Unique ID (e.g., User_88x2-99j).
- You are now "tagged" in Rubicon's domain, even though you never visited their site directly.

Basic Tracking Mechanisms

- Tracking requires:
 - (1) re-identifying a user.
 - (2) communicating id + visited site back to tracker.

```
▼ Hypertext Transfer Protocol
  ▶ GET /pixel/p-3aud4J6uA4Z6Y.gif?labels=InvisibleBox&busty=2710 HTTP/1.1\r\n
    Host: pixel.quantserve.com\r\n
    Connection: keep-alive\r\n
    Accept: image/webp,*/*;q=0.8\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_2) AppleWebKit/537.36
    Referer: http://www.theonion.com/\r\n
    Accept-Encoding: gzip,deflate,sdch\r\n
    Accept-Language: en-US,en;q=0.8\r\n
    Cookie: mc=52a65386-f1de1-00ade-0b26e; d=ENkBRgGHD4GYEA35MMIL74MKiyDs1A2MQI1Q
```

Anonymous Tracking Example (2)

- **Action:** You leave CNN and visit a completely different site, like Weather.com.
- **Mechanism:** Weather.com also uses Rubicon to sell ad space. When your browser loads the weather page, it again sends a request to Rubicon.
- **Result:** Your browser automatically sends the cookie back to Rubicon.
- Rubicon now knows that the same anonymous person who was just reading news is now checking the weather.

Anonymous Tracking Example (3)

- **Data Aggregation:** Over time, Rubicon sees User_88x2-99j on:
 - A tech blog (Interest: Computer Security)
 - An airline site (Interest: Travelling)
 - A fitness forum (Interest: Running)
 - ...and much more
- Rubicon don't know your name, but they have completed user profiling: they know your interest in every aspect, even your daily scheduling.

Anonymous Tracking Example (4)

- **Auction:** The next time you load a page, Rubicon holds a "Real-Time Bidding" (RTB) auction in milliseconds.
- They tell advertisers: "I have a user here who likes... and has the following habit: ... Who wants to show them an ad?"
- You see a highly specific ad for a "Developer Conference in London."

Tracking Technologies

- HTTP Cookies
- HTTP Auth
- HTTP Etags
- Content cache
- IE userData
- HTML5 protocol and content handlers
- HTML5 storage
- Flash cookies
- Silverlight storage
- TLS session ID & resume
- Browsing history
- window.name
- HTTP STS
- DNS cache
- “Zombie” cookies that respawn
(<http://samy.pl/evercookie>)

Fingerprinting Web Browsers

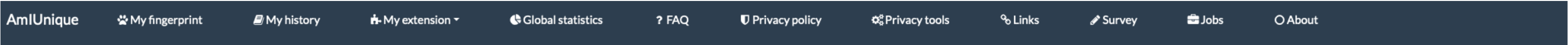
- User agent
- HTTP ACCEPT headers
- Browser plug-ins
- MIME support
- Clock skew
- Natural ordering of properties
- Installed fonts
- Cookies enabled?
- Browser add-ons
- Screen resolution
- HTML5 canvas (differences in graphics SW/HW!)
- Math constants from JS Engines





<https://panopticlick.eff.org>

Your browser fingerprint appears to be unique among the 229,038 tested in the past 45 days.



Learn how identifiable you are on the Internet

Help us investigate the diversity of web browsers.

This website aims at studying the diversity of browser fingerprints and providing developers with data to help them design good defenses. Contribute to the efforts by viewing your own browser fingerprint or consult the current statistics of data provided by users around the world!

View my browser fingerprint

If you click on this button, we will collect your browser fingerprint, we will put a cookie on your browser for a period of 4 months. More details are available in the privacy policy

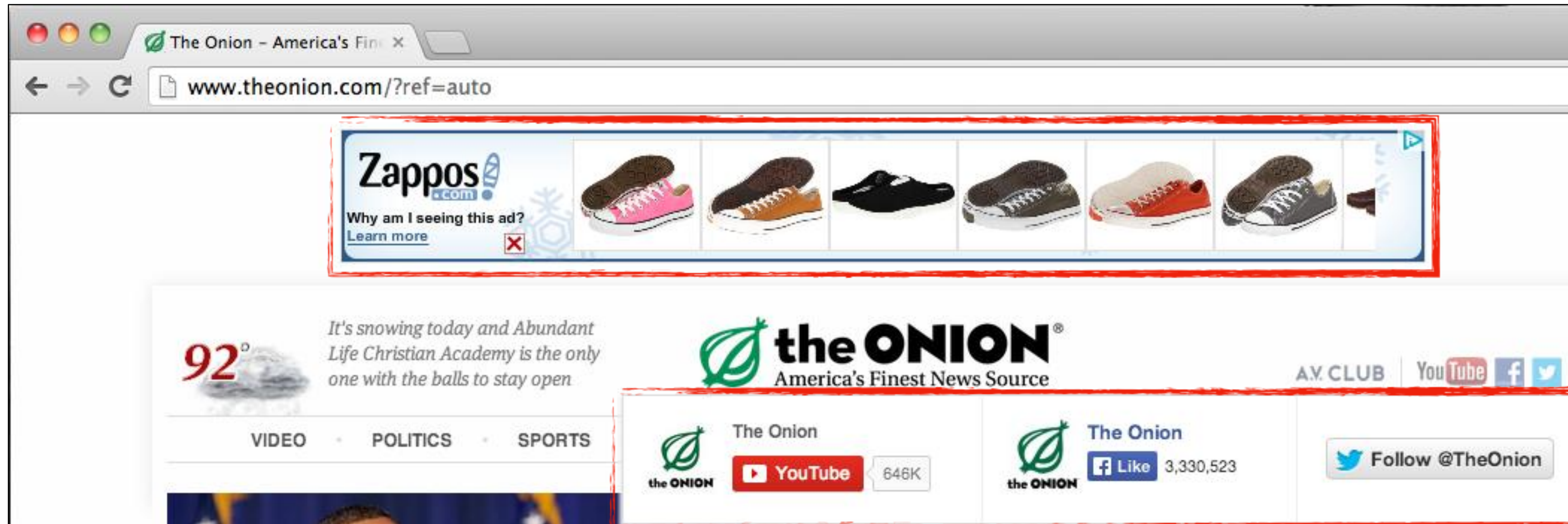
<https://amiunique.org>

You are unique among the 2,653,699 fingerprints in our entire dataset.

My Desktop Fingerprint

Proximity sensor ⓘ	32.14%	false
Keyboard layout ⓘ	21.58%	Not supported
Battery ⓘ	18.93%	Not supported
Connection ⓘ	16.92%	Not supported
key ⓘ	37.71%	cookie
Location bar ⓘ	36.47%	Visible
Menu bar ⓘ	36.49%	Visible
Personal bar ⓘ	36.46%	Visible
Status bar ⓘ	36.47%	Visible
Tool bar ⓘ	36.46%	Visible
Result state ⓘ	29.01%	No value
List of fonts (Flash) ⓘ	67.71%	Flash not detected
Screen resolution (Flash) ⓘ	68.81%	Flash not detected
Language (Flash) ⓘ	68.81%	Flash not detected
Platform (Flash) ⓘ	68.81%	Flash not detected

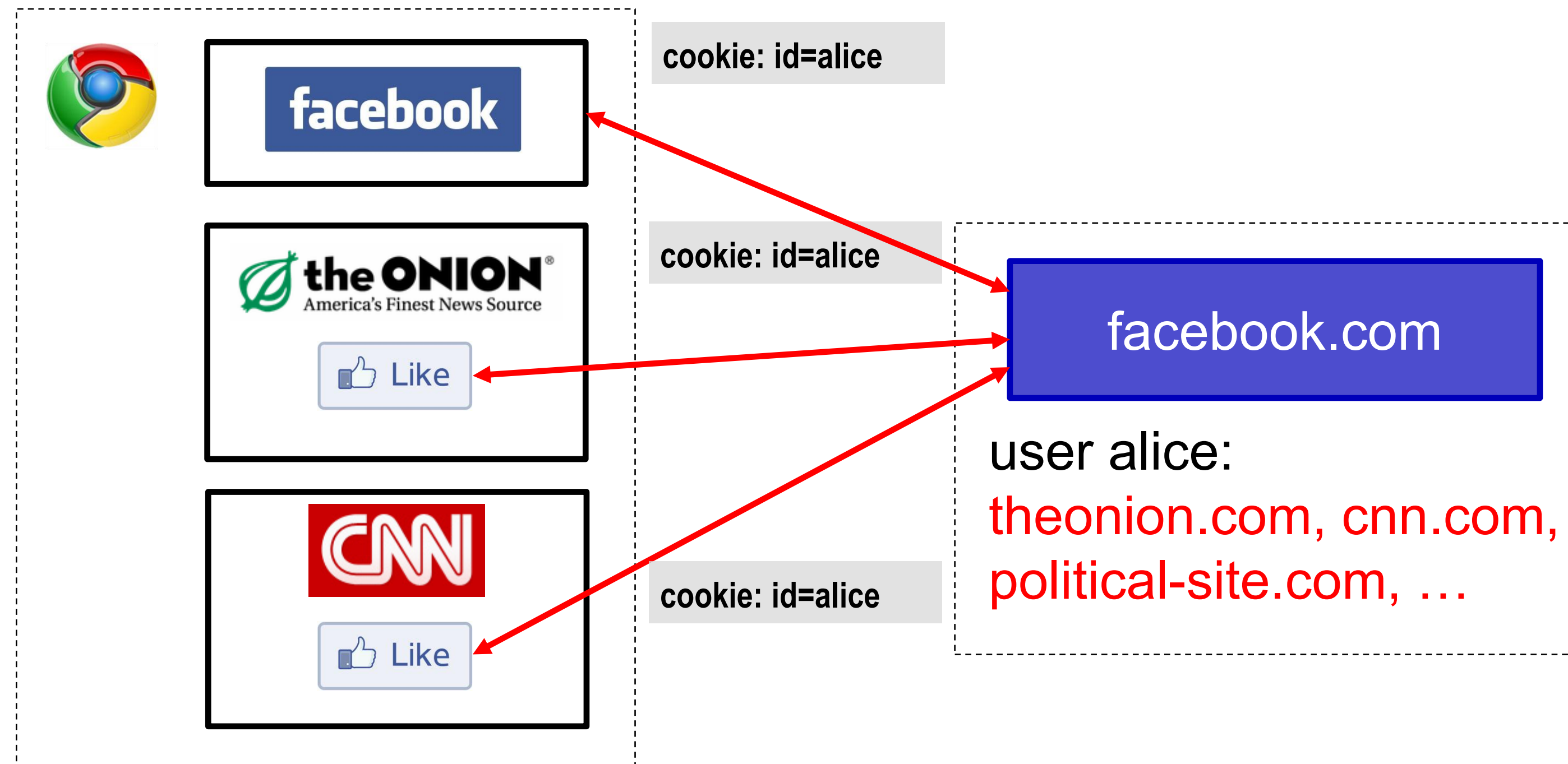
Other Trackers?



“Personal” Trackers



Personal Tracking



- Tracking is **not anonymous** (linked to accounts).
- Users **directly visit tracker's site** → evades some defenses.

Measuring Web Tracking

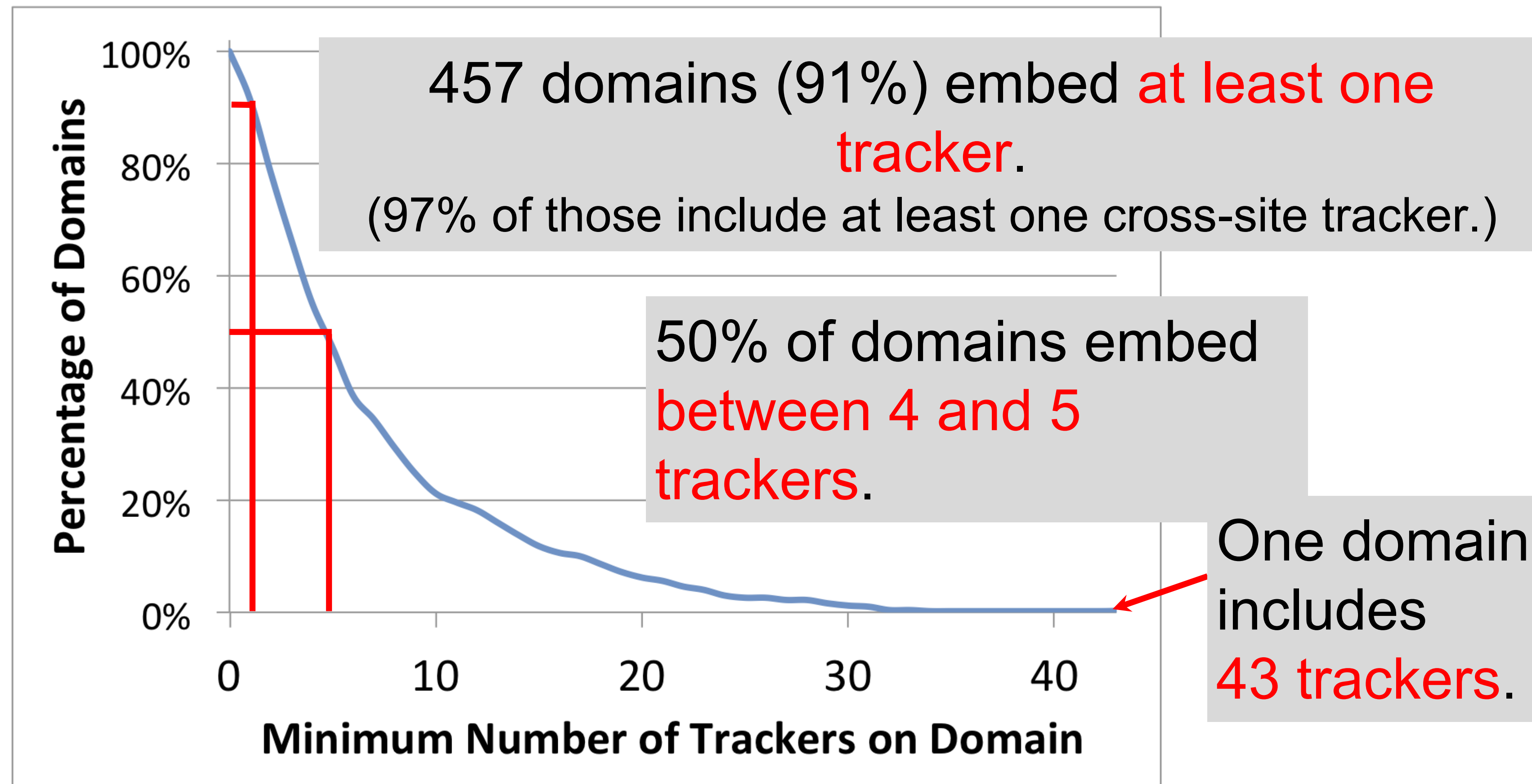
Measurement Study

- **Questions:**
 - How **prevalent** is tracking (of different types)?
 - How much of a user's browsing history is captured?
 - How effective are **defenses**?
- **Approach:** Build tool to **automatically crawl web, detect and categorize trackers** based on our taxonomy.

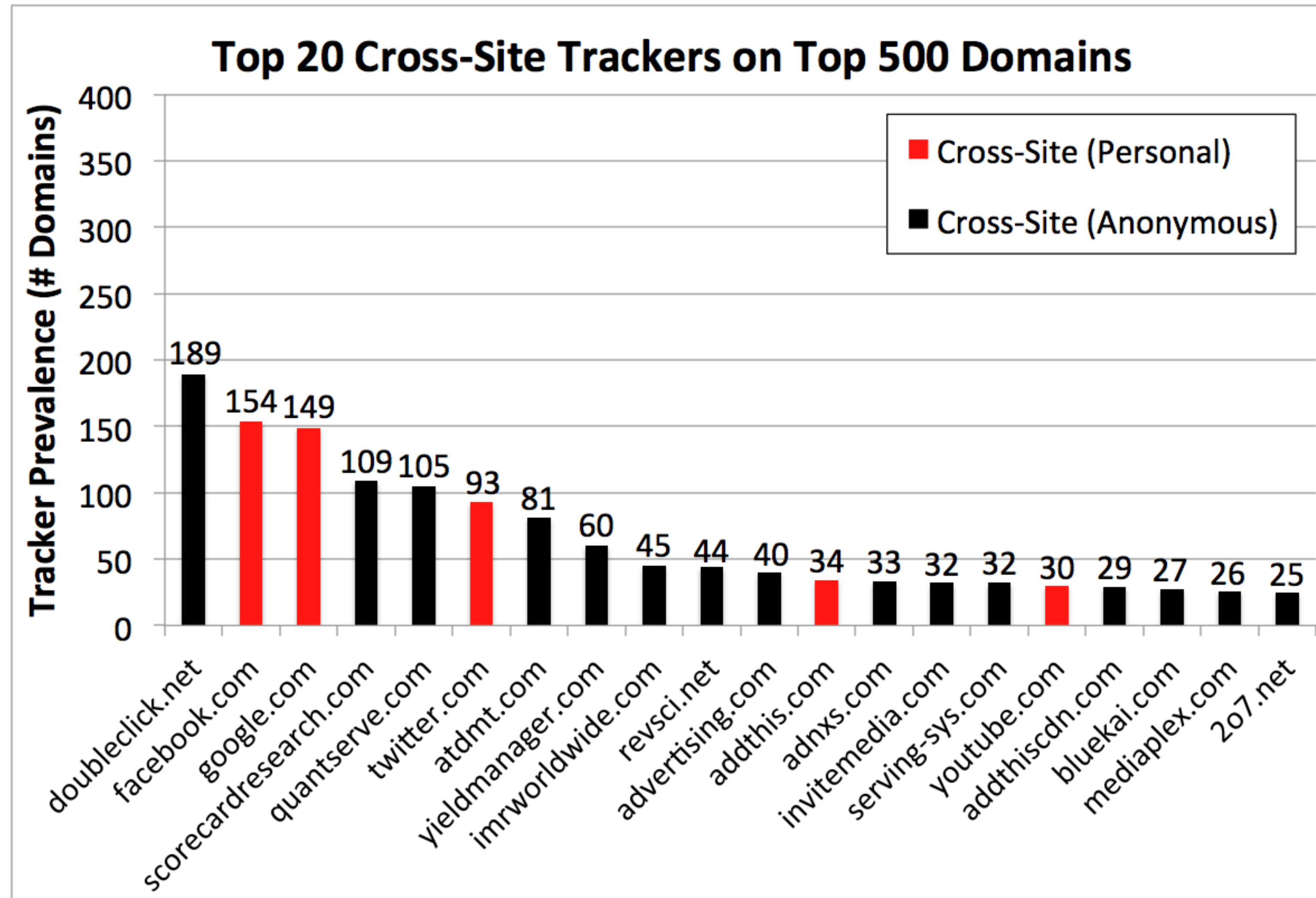
Longitudinal studies since then: **tracking has increased and become more complex.**

How prevalent is tracking? (2011)

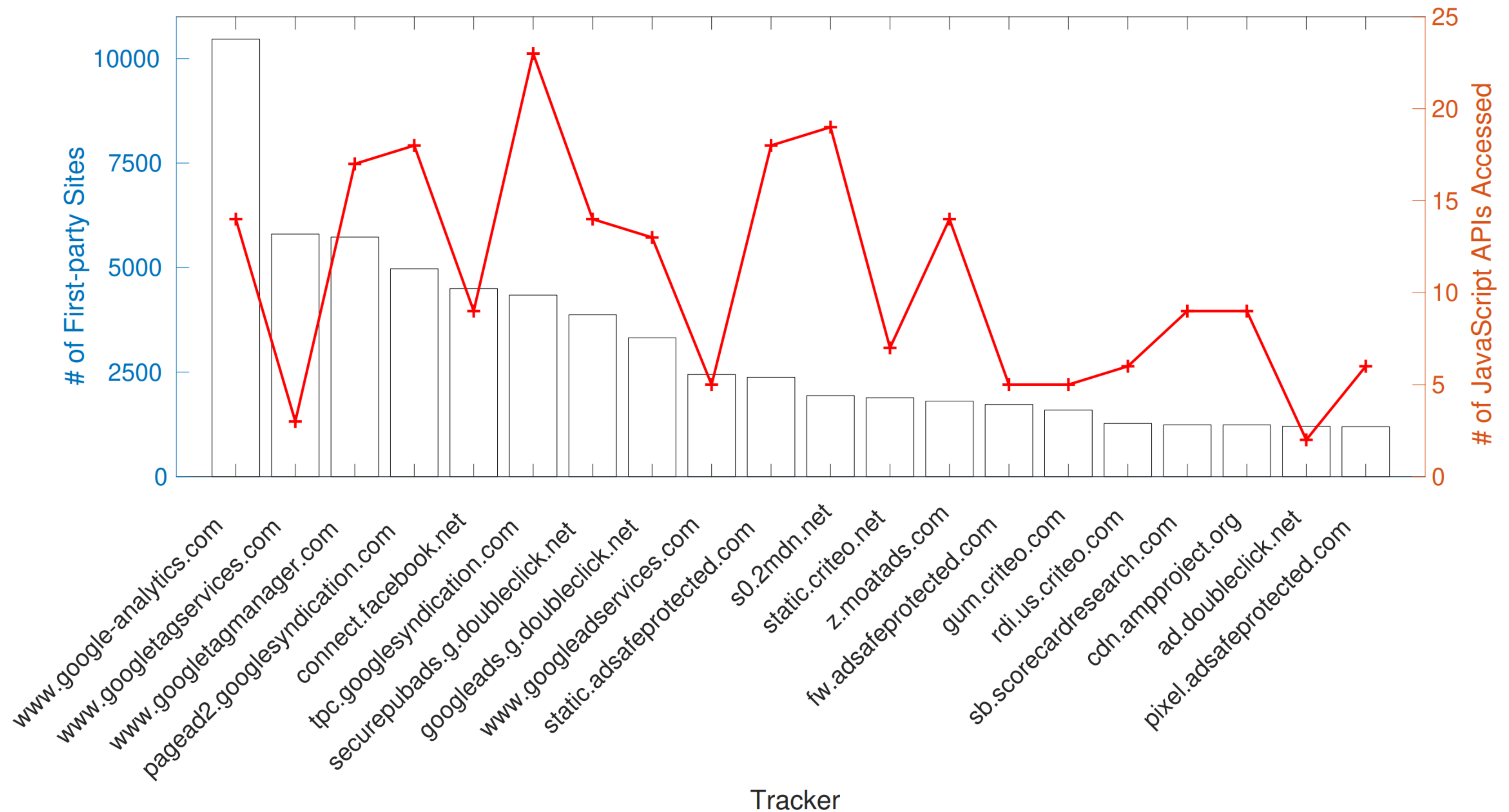
524 unique trackers on Alexa top 500 websites (homepages+ 4 links)



Who/what are the top trackers? (2011)



Who/what are the top trackers? (2020)



How has this changed over time?

- The web has existed for a while now...
 - What about tracking before 2011?
 - What about tracking before 2009?
- Solution: time travel!



**Internet Jones and the Raiders of the Lost Trackers:
An Archaeological Study of Web Tracking from 1996 to 2016**

Adam Lerner*, Anna Kornfeld Simpson*, Tadayoshi Kohno, Franziska Roesner
University of Washington
{lerner, aksimpso, yoshi, franzi}@cs.washington.edu

The Wayback Machine to the Rescue

INTERNET ARCHIVE

Wayback Machine

415 captures

18 Oct 2006 – 19 Sep 2020

http://cs.stonybrook.edu/

Go

OCT

DEC

JAN

06

2006

2007

Computer Science at Stony Brook University



Admission Research Facilities People Undergrad Graduate Events About Us

Welcome to the Computer Science Department at Stony Brook University

Established in 1969, the Computer Science Department at Stony Brook University is ranked consistently among the top quarter of Computer Science research departments in North America. In a recent Gourman report Stony Brook's undergraduate Computer Science program was ranked 15th nationwide and 2nd in New York State.

The department is the largest unit in the College of Engineering and Applied Sciences and is among the largest on the campus. Our faculty and students work closely together in an open, collegial atmosphere. The department is active in many of the major [research areas](#) in computer science with specialization in Visual Computing, Computer Systems, Networking and Security, Databases, Logic Programming and Deductive Systems, Concurrency and Verification, Algorithms and Complexity, and Computer Science Education. Our department is the primary participant in the [Center of Excellence in Wireless and Information Technology](#), a \$230 Million High-Tech Center at Stony Brook and one of only [five](#) in New York state. Our Computer Science major has been accredited by [ABET](#) , effective October 1, 2004.

We are located in the Computer Science Building at the center of the tree-lined Stony Brook campus in the beautiful village of Stony Brook, New York, a residential neighborhood 50 miles east of New York City on the north shore of Long Island. Plans are currently being drawn for a [new Computer Science building](#) adjacent to our current building, and for a [new building for Center of Excellence in Wireless and Information Technology](#).



Senator Charles E. Schumer visits Computer Science Department

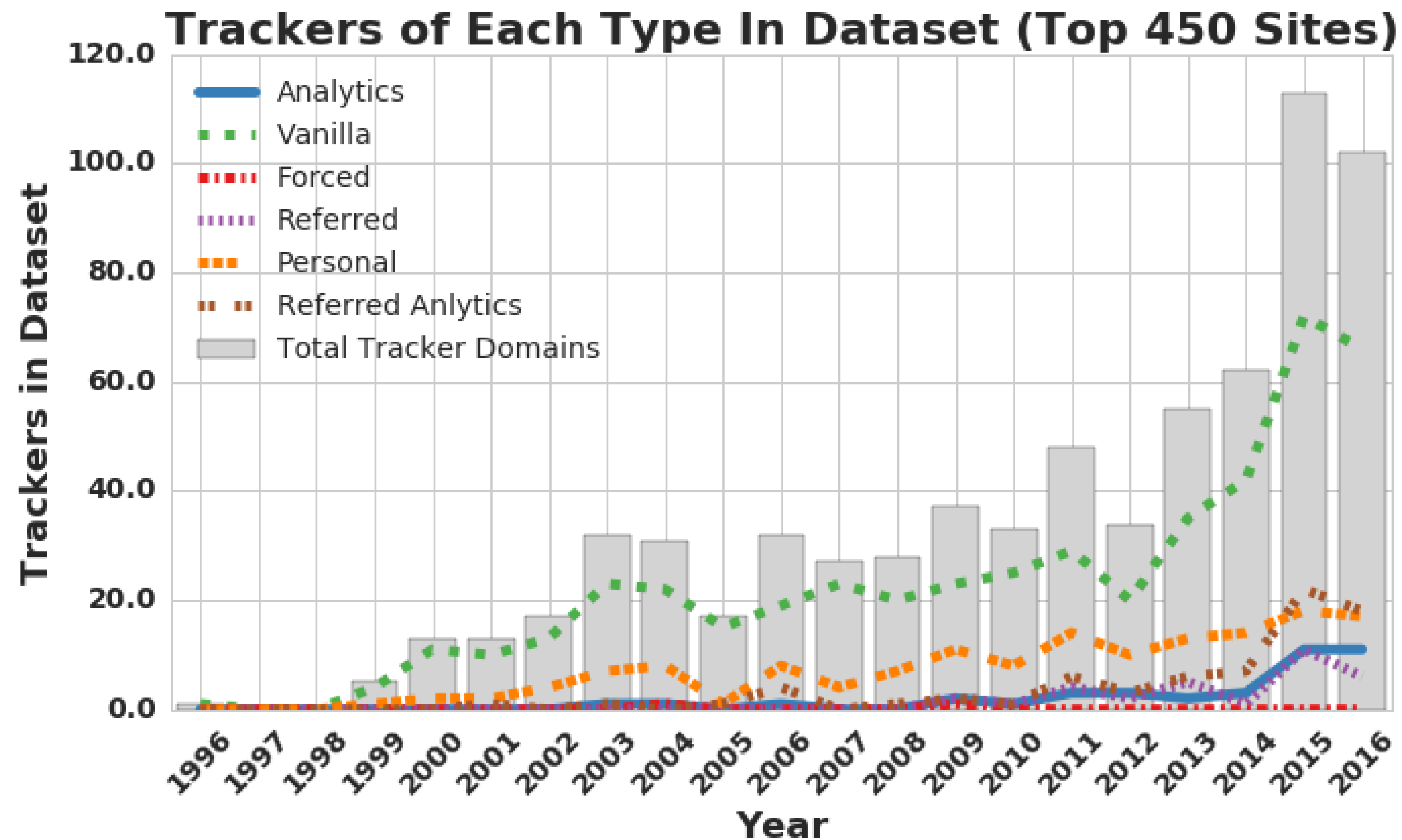
Open Faculty Positions



Alumni Registration

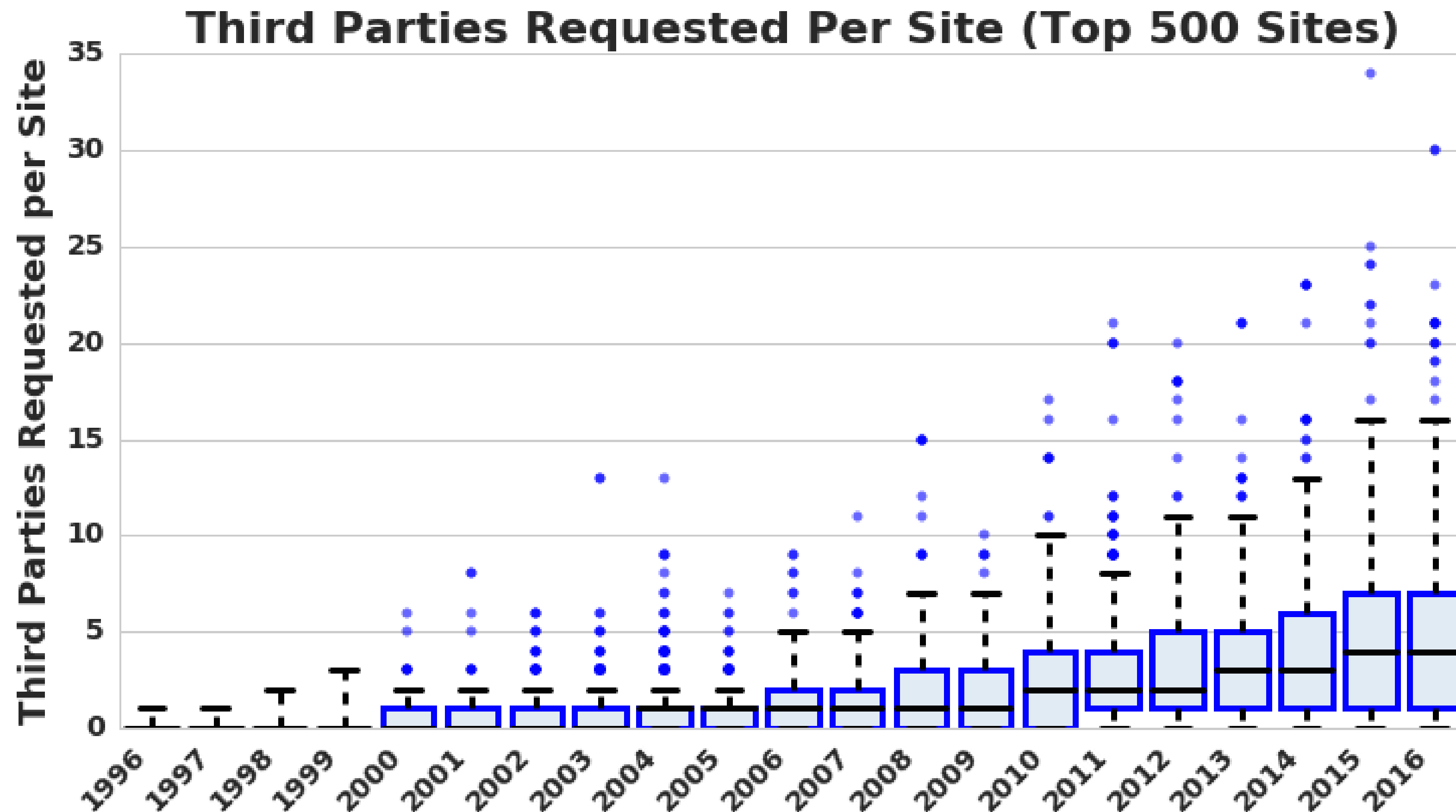
On May 10th, 2005, the Computer Science Department celebrated 35 years of education and innovation.

1996-2016: More & More Tracking



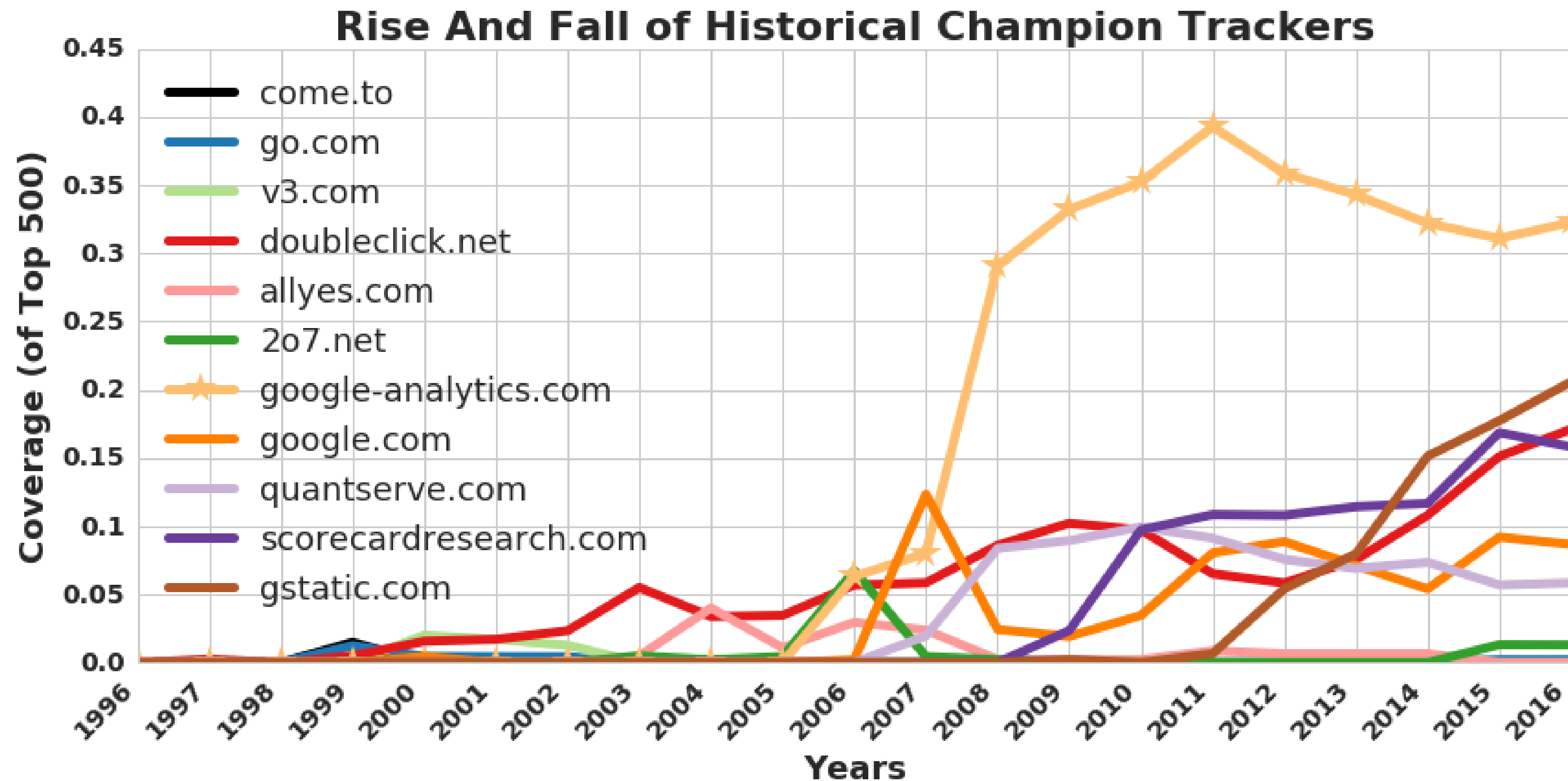
More trackers of more types

1996-2016: More & More Tracking



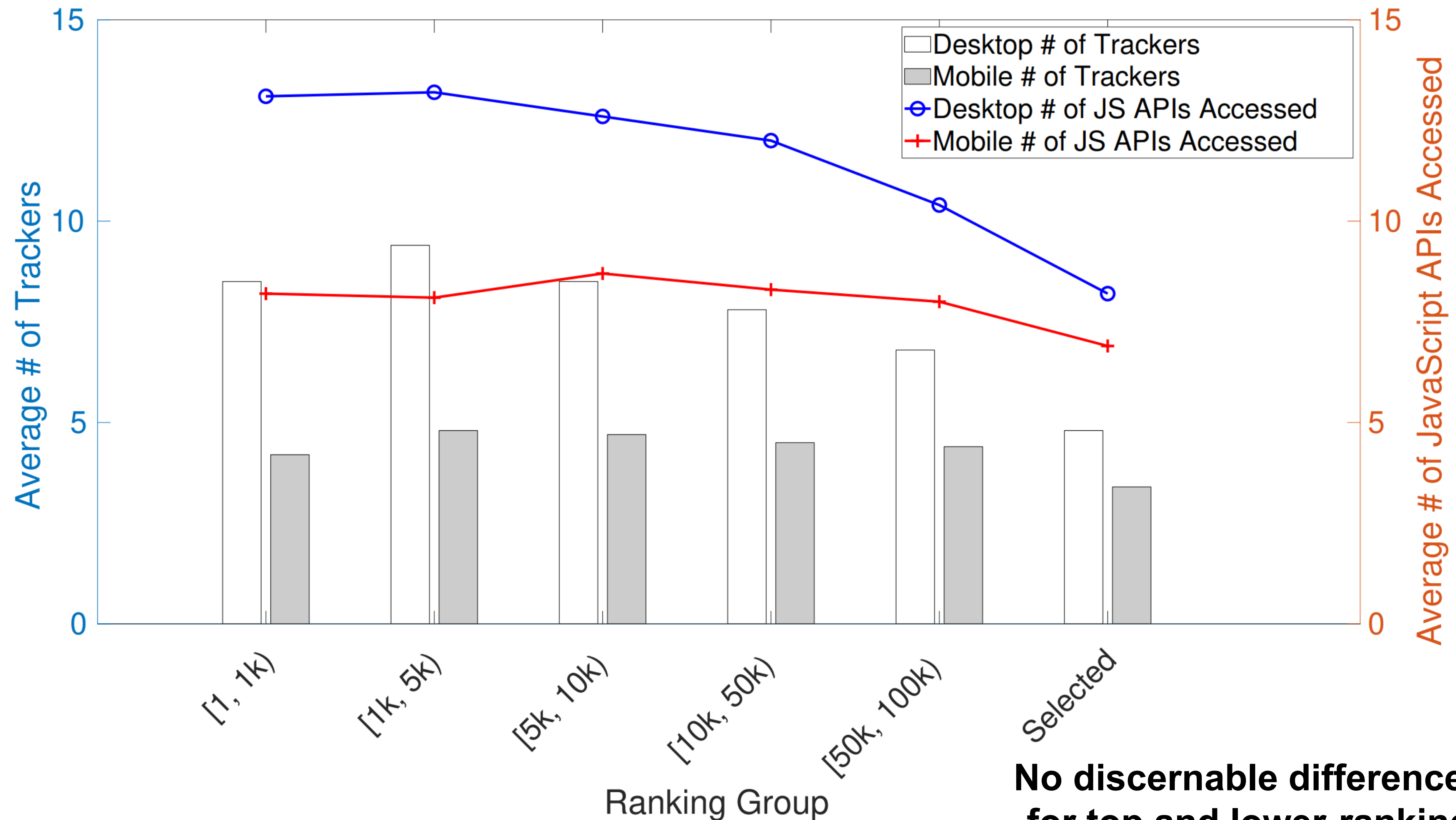
More trackers of more types, more per site

1996-2016: More & More Tracking



More trackers of more types, more per site, more coverage

Block Me If You Can (2022)



No discernable difference in tracking for top and lower-ranking websites.

Defenses to Reduce Tracking

Do Not Track (2010)

☒ Send a 'Do Not Track' request with your browsing traffic

Do Not Track is not a technical defense:
trackers must honor the request.

Private browsing mode

You've gone incognito

Now you can browse privately, and other people who use this device won't see your activity. However, downloads and bookmarks will be saved. [Learn more](#)

Chrome won't save the following information:

- Your browsing history
- Cookies and site data
- Information entered in forms

Your activity might still be visible to:

- Websites you visit
- Your employer or school
- Your internet service provider

Private browsing mode does not
save cookie. However, they are still
active within a session

Defenses to Reduce Tracking

Browser add-ons

Often rely on blacklists, which may be incomplete.



“uses algorithmic methods to decide what is and isn't tracking”



Tracking blockers may become trackers themselves!

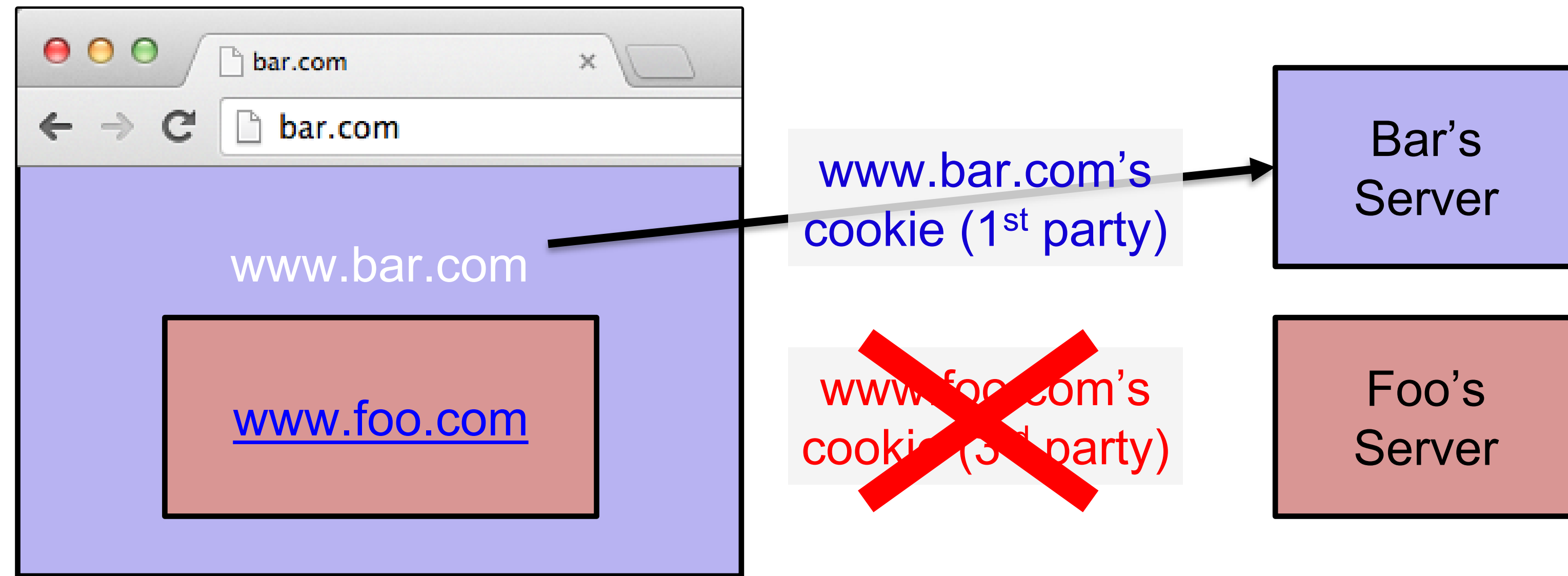
Previously, Ghostery sells its blocked list of trackers back to the advertising industry.



Look for open-source adblockers if you really need one.

Defenses to Reduce Tracking

Third-party cookie blocking



Websites began prominently asking for cookie consent, largely due to the enforcement of the European Union's General Data Protection Regulation (GDPR) in May 2018, which requires explicit consent for processing personal data, including data collected through cookies.

Quirks of 3rd Party Cookie Blocking

Cookies

- ☒ Allow local data to be set (recommended)
- ☐ Keep local data only until I quit my browser
- ☐ Block sites from setting any data
- ☒ Block third-party cookies and site data

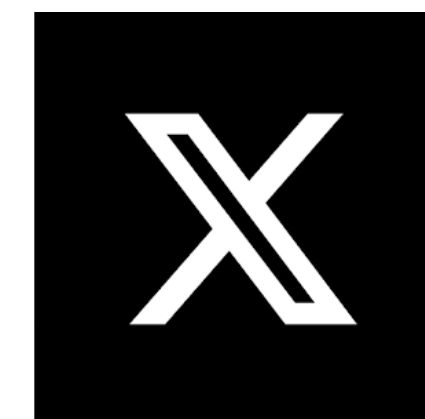
[Manage exceptions...](#) [All cookies and site data...](#)

In some browsers, this option means third-party cookies cannot be set, but they CAN be sent.

So if a third-party cookie is somehow set, **it can be used.**

How to get a cookie set?

One way: be a first party.



etc.

Defenses to Reduce Tracking

Browsers getting involved



Safari prevents trackers from following you across websites.

[Show Less](#)

Cross-site trackers

Some websites allow data collection companies called trackers to track your browsing activity. Trackers can follow you across multiple websites and combine your activity into a profile for advertisers.

Intelligent Tracking Prevention

Intelligent Tracking Prevention uses on-device machine learning to identify trackers and blocks them from accessing identifying information. Known trackers are independently identified by DuckDuckGo.

Last 30 days

Trackers prevented from profiling you

174

Websites that contacted trackers

64%

Safari and Firefox:

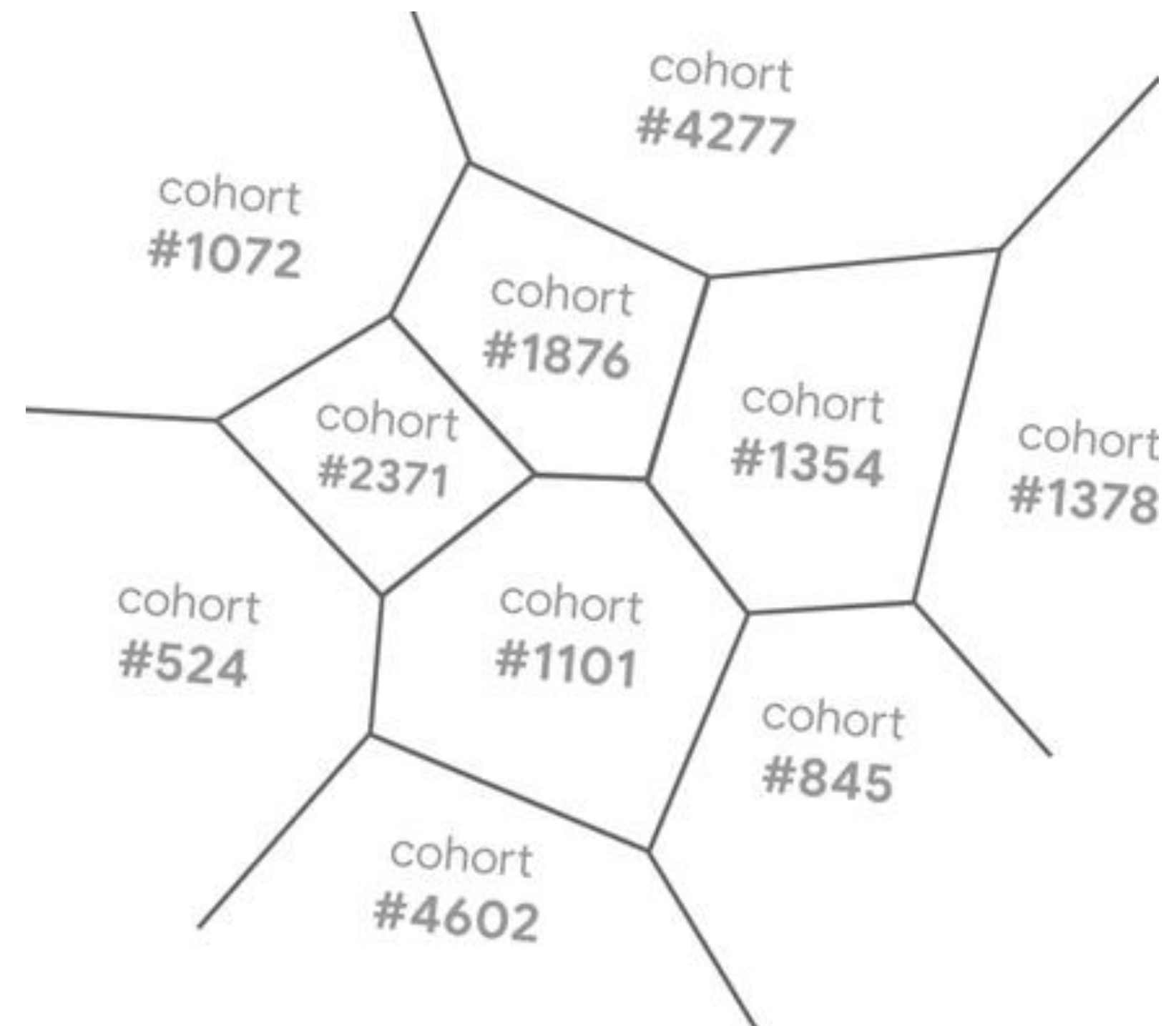
- These browsers already blocked third-party cookies years ago.
- You can still turn them on, but few bothers.
- If you use these browsers, the "death of 3rd party cookies" happened for you in 2017–2019.

Google's Plan: Privacy Sandbox

- Block all third-party cookies
- Federated Learning of Cohorts (FLOC)

There was a time when third-party cookies seemed to be gone forever.

...Until the plan was cancelled in 2024, partly because it will give Google an unfair advantage.



Defenses to Reduce Tracking

Policy Solution: General Data Protection Regulation (GDPR) 2018

- This regulation is the reason you see the “cookie consent” when you first visit a website.
- The European Union is behind the regulation.
- You may know that iPhone switched to USB-C also because of the EU regulations.

